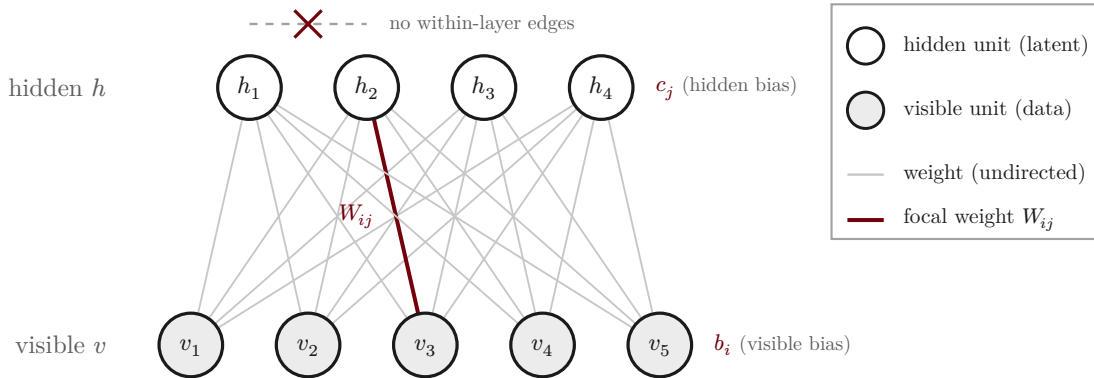


# Restricted Boltzmann Machine

bipartite undirected graph: every visible–hidden pair coupled by a symmetric weight  $W_{ij}$



$$E(v, h) = -b^\top v - c^\top h - v^\top W h, \quad p(v, h) = e^{-E(v, h)} / Z$$

restriction  $\Rightarrow$  factorial conditionals:  $p(h | v) = \prod_j p(h_j | v)$ ,  $p(v | h) = \prod_i p(v_i | h)$